Neelanshu Shukla

Gen-AI Data Science Solution Architect

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SUMMARY

Accomplished AI/ML and MLOps Architect with 12+ years of experience designing and deploying scalable solutions across industries including CPG, Retail, Banking, Insurance, BFSI, Pharma, Consulting, Supply chain, Manufacturing and Telecom. Specialised in Data Science, Generative AI, LLMs (RAG, GraphRAG), AI agents (Phi, Crew AI, LangGraph), and GenAI technologies. Proven expertise in developing Data Science, LLM and AI Agent backed models building end-to-end ML pipelines, integrating AI agents with enterprise systems, Cloud practices(Azure and AWS) and leading cross-functional teams to drive innovation, efficiency, and business value.

SKILLS

Technical Skills

Python | MLOps | Machine Learning | Deep Learning | SQL | API Development | Training & Download Pipeline | LLM"s | NLP | Flask API | Fast API | Hugging Face | GenAI | RAG | Transformer | Encoder/Decoder | Lang Chain | Phidata Framework | AI Agent & Tools | Whisper Model | API integration as third party agent or tool | GraphRAG | LangGraph | Open AI Model | Groq Model | Claude Model | LLAMA model | LLAMA Agent | Crew AI | AutoGen | LLAMA Index | AutoGPT | N8N | Temporal.io | Crew Orchestration | Eraser Architectural Design tool | CAG | Data Science | Python | Pytorch | Tensorflow | Keras

DevOps Tools

AWS | Azure | AWS GitHub | Azure Repo | Azure DevOps | Azure Monitor | Docker | Kubernetes | Jenkins | Azure SQL DB | Azure SDK | AWS EC2 | Snowflake | CI/CD Pipelines | MLFlow | AWS Cloudwatch | MLFlow | Groqcloud | EKS | Feature store | Azure DAB | Service Principle | Azure AD | Databricks Asset Bundle | Unity catalog | Data bricks Workflow | ADLS | Terraform | Datalake | Lake House Monitoring

Analytical Tools

AWS Sagemaker | VS Code | MS Excel | Power BI | Power Point | Azure ML Studio | Open AI | Azure Document Intelligence

WORK EXPERIENCE

Associate Principal Gen-AI & Data Science — Apr 2024 - Present

LTIMindtree, Bengaluru India

- Developed LLM backed product(ARGT) for EY Global Consulting to optimise solution design and workforce management with one form submission.
- Designed and deployed an end-to-end AI system using LangGraph, Crew AI, and DataBricks to interpret natural language sales queries from management, generate dynamic SQL for insurance datasets, visualise insights through automated dashboards, and deliver reports via email—combining LLM orchestration, multi-agent collaboration, and production-ready deployment on Azure for insurance client ERGO.

Lead Consultant MLOps — Nov 2021 - Mar 2024

HCL Technologies Ltd., Bengaluru, India

- Implemented MLOps and DLOps solutions for FedEx, addressing shipment delays and reducing operational expenses by 3% (3.5 million $).
- Utilised AWS services for deploying classification and time series models, improving Expected Delivery Date (EDD) accuracy.
- Enhanced Mastercard's Credit Risk Model and Business Identity Program, and optimised NSF handling for recurring payments, saved over (2.7 million $) card fees to network provider (AMEX, Discovery, VISA, Mastercard).

Consultant Machine Learning and AI — Aug 2020 - Oct 2021

Larsen & Toubro infotech, Bengaluru India

- Strategised and implemented solutions for AMESA market expansion at PepsiCo, optimising market strategies.
- Collaborated with Johnson & Johnson's R&D teams to develop a comprehensive data-driven framework for budget restructuring and bring down forecast vs actual gap from 15% to 7%.
- Enhanced efficiency in pharmaceutical development by encompassing resource estimation, research complexity, budget planning, and drug success rate modelling.

Advisory Analytics Expert — May 2019 - Jul 2020

Accenture Consulting , Bengaluru India

- Led a 6-member team for HP's global B2C & B2B e-commerce analytics, driving product catalog optimization, campaign automation (RFM, churn, cross-sell), and recommendation systems—resulting in $180M+ in digital campaign revenue.

Senior Data Scientist — Jul 2018 - Apr 2019

Tata Consultancy Services, Bengaluru, India

- Built a High Propensity Customer Model for Dick's Sporting Goods on GCP, achieving 25% conversion among active users and reducing campaign costs through targeted, gender- and age-based CRM strategies.

Assistant Manager Data Science — Jun 2013 - Jun 2018

Reliance Jio, Bengaluru, India

- Developed predictive models for various Jio products, enhancing revenue generation and strategic decision-making.
- Enhanced CRM effectiveness and optimized marketing spend with tailored campaigns targeting specific gender and age groups.

EDUCATION

B.Tech: Electronics and Communication
Faculty Of Engineering and Technology(Gurukul Kangri University)
71%
Jul 2009 - Jun 2013

PG Diploma: Data Science and Machine Learning
IIIT Bengaluru
Grade: 9
Jun 2017 - May 2018

CERTIFICATIONS

Azure Architect
Apr 2023 - Jun 2023

AWS Analytics Associate
Oct 2020 - Dec 2025

ACCOMPLISHMENTS

Best Analytics Leader awards by FedEx for supply chain resolution for new age future forward project.
Gen AI Leader award by LTIMindtree 2025 Global recognition for young leaders in AI adoptation.

PROJECTS

AMOD Automation — Mar 2025 - Present

AMOD(Automated Model Oriented Delivery) stands for complete start to end automated delivery where the Insurance product pipeline uses LLM model, AI Agents(NLP based agent, SQL query Agent, Dashboard Agent and Email Manager Agent) to generate the final product performance report and share it with stakeholder's on weekly and monthly basis, We have used Langgraph community, Lang Chain for state control and orchestration, CREW AI (Uses Crew agents as node functions inside LangGraph for best modularity) , Databricks, Azure lakehouse, Azure DevOps, DAB cookie cutter, Azure End Point Deployment.

ARGT- Auto response generation tool — May 2024 - Feb 2025

I played a pivotal role in helping EY Global Consulting develop a GenAI-powered product to optimise solution design and workforce management. The system captured customer requirements via intuitive UI forms, processed by a multimodal LLM to identify optimal solutions for business problems. I built end-to-end AI pipelines managing the entire model lifecycle—from development to deployment—using Azure DevOps, VS Code, Azure ML Studio, Azure SQL Database, AKS, and Azure OpenAI. Robust APIs (GET/POST) enabled seamless application interaction, six multi-class classification endpoints were integrated into the ARGT system as internal resources to provide use-case specific key notes. This significantly enhanced EY's ability to deliver tailored solutions and workforce planning for global partners. The project streamlined operations and improved decision-making, demonstrating my expertise in advanced AI and MLOps to drive business value.

BUV -Business Unit verification & NSF (Non settling fund) forecasting — Apr 2023 - Apr 2024

As an AI/ML Consultant for Mastercard's Banking initiatives, I strengthened the Credit Risk Model and Business Identity verification Program for Open Banking for Business (OBB) using integrating AWS services like S3, SageMaker, and EC2 Instances. For Open Banking for Customer (OBC), I optimised Non-Settling Fund AI models using AWS platform and monitor the performance via CloudWatch, enhancing payment processes and customer satisfaction. These initiatives advanced Mastercard's open banking capabilities with robust, AI-driven solutions, integrating GitLab CI/CD with AWS SDK, Jenkins and terraform for seamless deployment and management.

Future Forward — Dec 2021 - Mar 2023

As Management consultant for FedEx's Future Forward project, I implemented MLOps and DLOps solutions leveraging Azure Databricks, Azure DevOps, MLflow, Azure DAB, Data Bricks Workflow, CI/CD automation. FedEx faced operational challenges including delays, misplacements, and high service calls. Using HCL's Cascade program and Azure Databricks tools, I have developed and deployed supervised classification models and historical data-driven Stochastic Time Series and LSTM and Transformer based semantic and contextual search models to improve Expected Delivery Date (EDD) accuracy. A Resource Management initiative utilised supervised regression models to optimise service agent allocation, reducing call volumes and enhancing customer satisfaction, thereby cutting operational expenses while maintaining service excellence.

Budget Restructuring and drug trial research — Apr 2021 - Nov 2021

Collaborated with Johnson & Johnson's R&D teams, I have orchestrated a comprehensive data-driven framework encompassing resource estimation, research complexity, budget planning, and drug success rate modelling to enhance efficiency in pharmaceutical development. Multi-class classification and Regression model were developed for Site manager complexity, Programmer complexity and bio-statistician. For next project named drug trail research, Image classification CNN, YOLO deep learning model were design and developed.

Market Expansion AMESA — Aug 2020 - Mar 2021

As Data Science consultant we developed solutions to priorities cluster selection in AMESA region to strengthen the PEPSICO CSD sales and also manages the retail supplies and stocks. we have developed models for priority region selection based on POI's and foot fall with GPS coordinate, for supplies management we have design the latency supplies model to predict the stock inventory sell out time line.

Recommendation Engine for popular product and Up-sell — Jun 2019 - Jul 2020

Hewlett-Packard's B2B retail and e-commerce global business digital sale and procurement platform were managed by us, Our focus was on developing analytics- driven solutions for product management and customer catalog handling. The Digital Analytics team executes strategic digital campaigns aimed at boosting sales and revenue. We've successfully deployed market basket analysis for cross-selling and a recommendation engine, generating expected sales figures through live campaigns. Automation plays a crucial role, including product category mapping, RFM segmentation, churn modelling, and upsell models, all integrated with our digital initiatives.